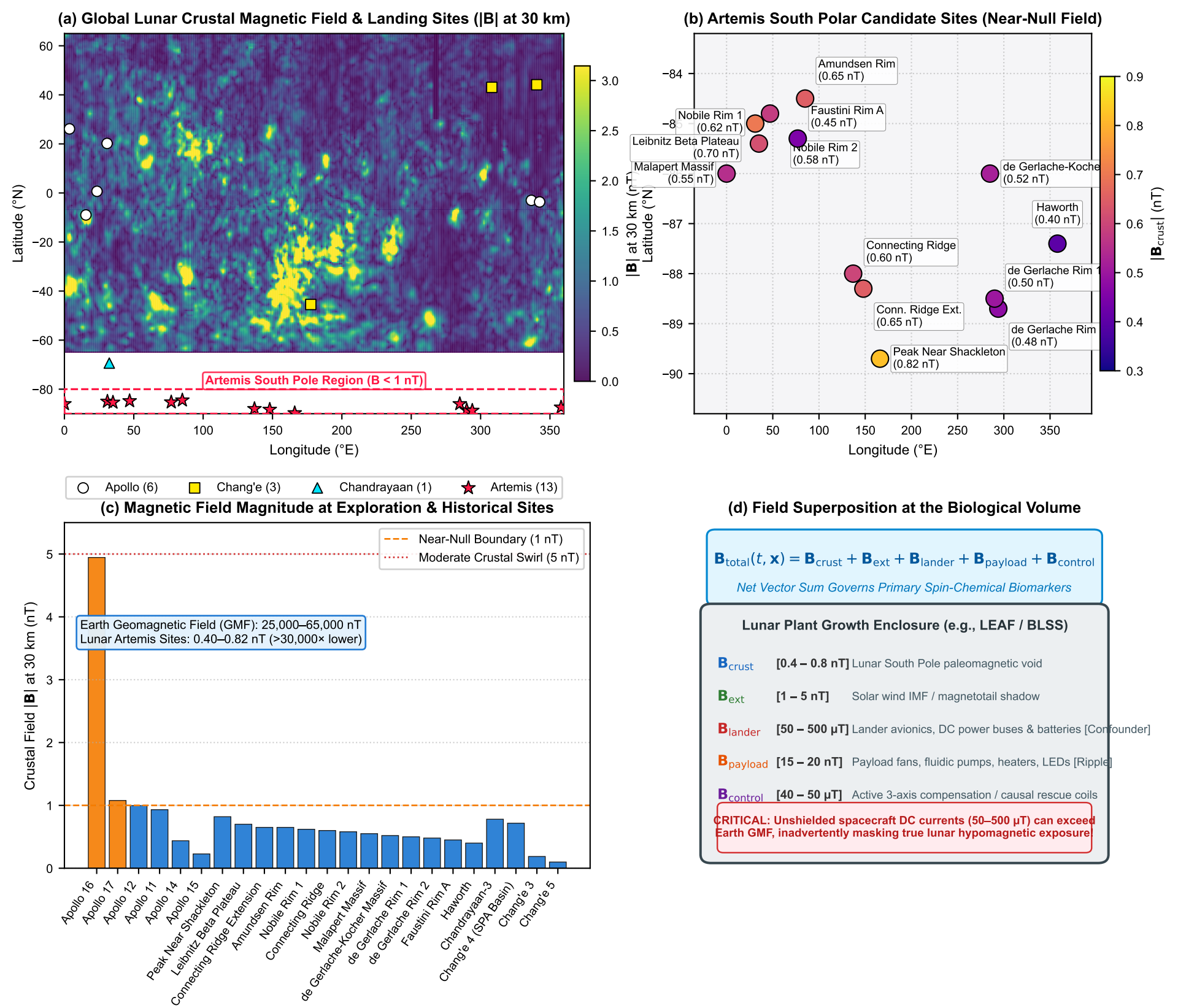

(a) Global Lunar Crustal Magnetic Field & Landing Sites (|B| at 30 km)
Latitude (°N)
Longitude (°E)
|B| at 30 km (nT)
Artemis South Pole Region (B < 1 nT)
Apollo (6)
Chang'e (3)
Chandrayaan (1)
Artemis (13)
(b) Artemis South Polar Candidate Sites (Near-Null Field)
Amundsen Rim (0.65 nT)
Faustini Rim A (0.45 nT)
Nobile Rim 1 (0.62 nT)
Leibnitz Beta Plateau (0.70 nT)
Nobile Rim 2 (0.58 nT)
Malapert Massif (0.55 nT)
de Gerlache-Koche (0.52 nT)
Haworth (0.40 nT)
Connecting Ridge (0.60 nT)
de Gerlache Rim 1 (0.50 nT)
Conn. Ridge Ext. (0.65 nT)
de Gerlache Rim (0.48 nT)
Peak Near Shackleton (0.82 nT)
|B_crust| (nT)
(c) Magnetic Field Magnitude at Exploration & Historical Sites
Near-Null Boundary (1 nT)
Moderate Crustal Swirl (5 nT)
Earth Geomagnetic Field (GMF): 25,000–65,000 nT
Lunar Artemis Sites: 0.40–0.82 nT (>30,000× lower)
Crustal Field |B| at 30 km (nT)
Apollo 16
Apollo 17
Apollo 12
Apollo 11
Apollo 14
Apollo 15
Peak Near Shackleton
Leibnitz Beta Plateau
Connecting Ridge Extension
Amundsen Rim
Nobile Rim 1
Connecting Ridge
Nobile Rim 2
Malapert Massif
de Gerlache-Kocher Massif
de Gerlache Rim 1
de Gerlache Rim 2
Faustini Rim A
Haworth
Chandrayaan-3
Chang'e 4 (SPA Basin)
Chang'e 3
Chang'e 5
(d) Field Superposition at the Biological Volume
$\mathbf{B}_{total}(t, \mathbf{x}) = \mathbf{B}_{crust} + \mathbf{B}_{ext} + \mathbf{B}_{lander} + \mathbf{B}_{payload} + \mathbf{B}_{control}$
Net Vector Sum Governs Primary Spin-Chemical Biomarkers
Lunar Plant Growth Enclosure (e.g., LEAF / BLSS)
$\mathbf{B}_{crust}$ [0.4 – 0.8 nT] Lunar South Pole paleomagnetic void
$\mathbf{B}_{ext}$ [1 – 5 nT] Solar wind IMF / magnetotail shadow
$\mathbf{B}_{lander}$ [50 – 500 μT] Lander avionics, DC power buses & batteries [Confounder]
$\mathbf{B}_{payload}$ [15 – 20 nT] Payload fans, fluidic pumps, heaters, LEDs [Ripple]
$\mathbf{B}_{control}$ [40 – 50 μT] Active 3-axis compensation / causal rescue coils
CRITICAL: Unshielded spacecraft DC currents (50–500 μT) can exceed Earth GMF, inadvertently masking true lunar hypomagnetic exposure!